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SHEET MATERIAL DISPENSER ASSEMBLY FOR SELECTIVELY DISPENSING SHEET MATERIAL FROM A PLURALITY OF SUPPLIES OF ROLLED SHEET MATERIAL

Abstract

A dispenser assembly facilitating selective dispensing of sheet material from a plurality of supplies of sheet material can be provided. The dispenser assembly can include a drive system that includes a plurality of driven rollers configured to engage and move sheet material from a respective supply of sheet material and a drive mechanism configured to drive the plurality of driven rollers. When the drive mechanism is driven in one direction, one of the plurality of driven rollers is rotated to dispense sheet material from one of the plurality of supplies of sheet material, and when the drive mechanism is driven in the opposite direction, another one of the plurality of driven rollers is rotated to dispense sheet material from another supply of sheet material.

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Background/Summary

CROSS-REFERENCE [0001] The present Patent Application is a continuation of co-pending U.S. patent application Ser. No. 18/107,260, filed Feb. 8, 2023, which claims benefit of U.S. Provisional Patent Application No. 63/307,699, filed Feb. 8, 2022, and claims benefit of U.S. Provisional Patent Application No. 63/337,371, filed May 2, 2022, titled "SHEET MATERIAL DISPENSER ASSEMBLY FOR SELECTIVELY DISPENSING SHEET MATERIAL FROM A PLURALITY OF SUPPLIES OF ROLLED SHEET MATERIAL." INCORPORATION BY REFERENCE [0002] The disclosures and figures of U.S. patent application Ser. No. 18/107,260, filed Feb. 8, 2023, U.S. Provisional Patent Application No. 63/307,699, filed on Feb. 8, 2022, and U.S. Provisional Patent Application No. 63/337,371, filed May 2, 2022 are specifically incorporated by reference herein as if set forth in their entireties.

TECHNICAL FIELD
[0003] In one aspect, the present disclosure is directed to dispenser assemblies for rolled sheet materials, and more particularly, is directed to dispenser assemblies for selectively dispensing from a plurality of supplies of rolled sheet material. Other aspects are also described.
BACKGROUND
[0004] Dispensers for sheet materials, such as for dispensing tissue paper, paper towels, or other paper products, are commonly used in hospitals, restrooms, and other facilities. Some dispensers have more than one supply of sheet material, e.g., multiple rolls of sheet material, for dispensing/feeding. When a supply of sheet material in such dispensers is running low or has been fully dispensed, a transfer of the feeding of sheet material to a new supply generally must be performed, which often must be done manually. Accordingly, it can be seen that a need exists for a dispenser assembly that can selectively switch/transfer the feeding/dispensing of sheet material between a plurality of supplies of sheet material, e.g., when a supply of sheet material is running low or has been fully dispensed. The present disclosure addresses these and other related and unrelated problems/issues in the relevant art.
SUMMARY
[0005] In one aspect, the present disclosure is directed to a dispenser assembly for dispensing sheet materials such as rolls of tissue, paper towels, and/or other rolled sheet material products. The dispenser assembly generally includes a dispenser housing having a plurality of supplies of rolled sheet material supported therein.
[0006] Each supply of rolled sheet material is supported by a corresponding support assembly within the dispenser housing. In one construction, the plurality of supplies of sheet material can include a first supply of sheet material supported by a corresponding first support assembly, and a second supply of sheet material supported by a corresponding second support assembly. The first and second support assemblies can be spaced apart from each other along the dispenser housing.
[0007] The dispenser assembly further can include a dispensing system for controlling the dispensing of selected, predetermined amounts of sheet material from at least one of the plurality of supplies of sheet material. The dispensing system can include a plurality of driven roller assemblies for engaging and driving the sheet material from the supplies of rolled sheet material. Each driven roller assembly generally will be associated with at least one supply of the plurality of supplies of

sheet material for dispensing sheet material therefrom. For example, the first supply of rolled sheet material can be dispensed by a first driven roller assembly and the second supply of rolled sheet material can be dispensed by a second driven roller assembly.

[0008] Each driven roller assembly can have at least one driven roller driven by a drive mechanism (e.g., a motor or other suitable drive mechanism) in communication therewith. In one variation, the drive mechanism can be operatively connected to the driven roller(s) by a belt or series of belts (e.g., one or more belts engaging a belt pulley or belt gear connected to each of the driven rollers).

[0009] The dispensing assembly further can include at least one guide roller that engages the sheet material and is rotatable with the rotation of the driven roller to help facilitate feeding and dispensing of the sheet material. The dispenser assembly further can include additional guide or pressing rollers positioned adjacent each of the driven rollers to help guide the sheet material during dispensing thereof without departing from the scope of the present disclosure.

[0010] Each of the driven rollers can be configured to rotate in a desired or selected direction, and typically can be rotated by the drive mechanism for a selected number of rotations as needed to dispense the selected amounts of sheet material from their corresponding supply of rolled sheet material, but generally will remain stationary when the drive mechanism is reversed or driven in the opposite direction. For example, each driven roller can include or can be coupled to a clutch mechanism (e.g., a hybrid or one-way clutch mechanism) or other disengagable drive connection that engages the driven roller and causes it to rotate when driven/rotated in one direction and disengages the driven roller and allows it to stay substantially stationary when driven in the opposite direction.

[0011] For example, the first driven roller can be rotated when the drive mechanism is driven in a first direction to dispense sheet material from the first supply of rolled sheet material, while the second driven roller can remain generally stationary such that sheet material is not dispensed from the second supply of rolled sheet material. When the drive mechanism is driven in a second direction, the second driven roller can be rotated to dispense selected predetermined amounts of sheet material from the second supply of rolled sheet material, while the first driven roller can be disengaged and remain generally stationary such that sheet material is not dispensed therefrom.

[0012] Accordingly, the dispenser assembly of the present disclosure provides for selective dispensing of sheet material from the plurality of supplies of sheet material as needed. For example, upon a change or reversing of the driving direction of the drive mechanism, the dispenser can switch the dispensing of sheet material from the one supply of sheet material to the other. This change or switch/transfer of feeding from one supply to another can be substantially automatic, i.e., in response to a signal from a sensor or monitoring system, by a command from a control system for the dispenser, manually by a switch upon receipt of one or more signals from a device external to the dispenser assembly, etc.

[0013] The drive assembly additionally can include a tensioning assembly including one or more biasing members for providing a substantially constant tension along the drive belt. In one variation, the one or more biasing members (e.g., including one or more tension springs) can be operatively connected to the motor (e.g., one end of the one or more springs can be connected to the motor or a support therefor, and another end of the one or more springs can be connected to the dispenser housing or a component attached thereto).

[0014] The tensioning assembly can include a bracket movably supporting the drive mechanism along the dispenser housing, and the one or more biasing members can be coupled to the bracket to bias the tensioning assembly sufficient to apply tension along the drive belt and/or for providing dampening of vibrations from an operation of the dispenser assembly.

[0015] The dispenser assembly can include at least one cutting mechanism (e.g., including a tear bar(s), serrated cutting blade(s), knife(s), or other sharpened portion(s)) positioned along the discharge of the dispenser housing for severance of dispensed sheet material from the supplies of sheet material.

[0016] The dispenser assembly can include a pawl member assembly including a pivotally mounted pawl member located proximate or otherwise along the cutting mechanism such that movement of the sheet material into the cutting mechanism for severance thereof moves the pawl member from a first position to a second position. The pawl member assembly further can generate one or more signals that can be sent to a control circuit of the dispenser to notify the control circuit that a portion of the dispensed sheet material has been removed.
[0017] The dispensing assembly also can include a sheet material detection sensor including an emitter and a detector focused across at least a portion of the discharge path(s) extending through the discharge. The sheet material detection sensor can be activated by a control system of the dispenser assembly to verify that the sheet material has been removed from the discharge.
[0018] The dispensing assembly further can include a monitoring system configured to determine a supply level of the supplies of sheet material, and upon a determination that the supply level of the supplies of sheet material is below a threshold level, the direction of the drive mechanism can be changed.
[0019] Still more aspects and advantages of these embodiments and other embodiments, are discussed in detail herein. Moreover, it is to be understood that both the foregoing information and the following detailed description provide merely illustrative examples of various aspects and embodiments and are intended to provide an overview or framework for understanding the nature and character of the various aspects and embodiments disclosed herein. Accordingly, these and other objects, along with advantages and features of the present disclosure herein disclosed, will become apparent through reference to the following description and the accompanying drawings. Furthermore, it is to be understood that the features of the various embodiments described herein are not mutually exclusive and may exist in various combinations and permutations.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0020] The accompanying drawings, which are included to provide a further understanding of the embodiments of the present disclosure, are incorporated in and constitute a part of this specification, illustrate embodiments of the present disclosure, and together with the detailed description, serve to explain the principles of the embodiments discussed herein. No attempt is made to show structural details of this disclosure in more detail than can be necessary for a fundamental understanding of the exemplary embodiments discussed herein and the various ways in which they can be practiced. According to common practice, the various features of the drawings discussed below are not necessarily drawn to scale. Dimensions of various features and elements in the drawings can be expanded or reduced to more clearly illustrate the embodiments of the disclosure. The use of the same reference symbols in different drawings indicates similar or identical items.
[0021] FIG. **1** provides a schematic illustration of a dispensing assembly for selectively dispensing a predetermined amount of sheet material from a plurality of supplies of sheet material according to principles of the present disclosure.
[0022] FIG. **2**A shows a perspective view of a drive system for the dispensing assembly of FIG. **1**.
[0023] FIG. **2**B shows a belt pulley of a driven roller of the drive system of FIG. **2**A with an integrated clutch mechanism according to principles of the of the present disclosure.
[0024] FIG. **2**C illustrates a drive system according to further principles of the present disclosure.
[0025] FIGS. **3**A-**3**B illustrate examples of a cutting mechanism and pawl member according to example constructions of the present disclosure.
[0026] FIGS. **4**A-**4**B show perspective and cross-sectional views of a tensioning assembly according to principles of the present disclosure.

[0027] FIG. **5** shows a block diagram of an example of a control system of the dispenser assembly according to principles of the present disclosure.
[0028] FIGS. **6-8** are views of a dispenser assembly according to additional embodiments of the disclosure.
[0029] FIGS. **9-13** are views of a dispensing system of the dispenser assembly of FIGS. **6-8**.
[0030] FIGS. **14-18** are views of a transmission assembly of the dispensing system of FIGS. **9-13**.
[0031] FIGS. **19**A-**19**E show additional views of the features of FIGS. **6-18**.
[0032] FIGS. **20**A-**22** are views of a dispensing system according to additional embodiments of the disclosure.
[0033] FIGS. **23-27**B are views of a transmission assembly of the dispensing system of FIGS. **20**A-**22**.
[0034] FIGS. **28**A and **28**B show views of other embodiments of a dispensing system.
[0035] FIGS. **28**C and **28**D show views of other embodiments of a drive mechanism for a dispensing system.

DETAILED DESCRIPTION

[0036] The present invention can be understood more readily by reference to the following detailed description, examples, drawings, and claims, and their previous and following description. However, before the present devices, systems, and/or methods are disclosed and described, it is to be understood that this invention is not limited to the specific devices, systems, and/or methods disclosed unless otherwise specified, and, as such, can, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting.
[0037] The following description of the invention is provided as an enabling teaching of the invention in its best, currently known embodiment. To this end, those skilled in the relevant art will recognize and appreciate that many changes can be made to the various aspects of the invention described herein, while still obtaining the beneficial results of the present invention. It will also be apparent that some of the desired benefits of the present invention can be obtained by selecting some of the features of the present invention without utilizing other features. Accordingly, those who work in the art will recognize that many modifications and adaptations to the present invention are possible and can even be desirable in certain circumstances and are a part of the present invention. Thus, the following description is provided as illustrative of the principles of the present invention and not in limitation thereof.
[0038] As used throughout, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a driven roller” can include two or more such driven rollers unless the context indicates otherwise.
[0039] Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, another aspect includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another aspect. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.
[0040] As used herein, the terms “optional” or “optionally” mean that the subsequently described event or circumstance can or cannot occur, and that the description includes instances where said event or circumstance occurs and instances where it does not.
[0041] The word “or” as used herein means any one member of a particular list and also includes any combination of members of that list. Further, one should note that conditional language, such as, among others, “can,” “could,” “might,” or “can,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain aspects include, while other aspects do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are

in any way required for one or more particular aspects or that one or more particular aspects necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment.

[0042] The phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. As used herein, the term "plurality" refers to two or more items or components. The terms "comprising," "including," "carrying," "having," "containing," and "involving," whether in the written description or the claims and the like, are open-ended terms, i.e., to mean "including but not limited to." Thus, the use of such terms is meant to encompass the items listed thereafter, and equivalents thereof, as well as additional items. Only the transitional phrases "consisting of" and "consisting essentially of," are closed or semi-closed transitional phrases, respectively, with respect to any claims. Use of ordinal terms such as "first," "second," "third," and the like in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish claim elements.

[0043] Disclosed are components that can be used to perform the disclosed methods and systems. These and other components are disclosed herein, and it is understood that when combinations, subsets, interactions, groups, etc. of these components are disclosed that while specific reference to each various individual and collective combinations and permutation of these cannot be explicitly disclosed, each is specifically contemplated and described herein, for all methods and systems. This applies to all aspects of this application including, but not limited to, steps in disclosed methods. Thus, if there are a variety of additional steps that can be performed it is understood that each of these additional steps can be performed with any specific embodiment or combination of embodiments of the disclosed methods.

[0044] The present methods and systems can be understood more readily by reference to the following detailed description of preferred embodiments and the examples included therein and to the Figures and their previous and following description.

[0045] FIG. **1** shows a dispenser assembly **10** for dispensing a rolled sheet material **11**, such as tissue rolls, paper towel rolls, or other suitable rolled sheet material products. As shown in FIG. **1**, the dispenser assembly **10** can include a dispenser housing **12** having a cover **12**A that is movable/removable to allow access to the components of the dispenser assembly **10**, and a backing portion **12**B that is configured to mount or otherwise connect (e.g., via fasteners, adhesive, etc.) to the dispenser assembly **10** to a wall, partition, or other suitable support within a facility, such as, for example, a restroom, hospital room, and the like. The dispenser housing **12** can be formed from plastic materials, metallic materials, other suitable synthetic or composite materials, or combinations thereof. The dispenser housing **12** further includes one or more chambers or compartments **13** defined therein and sized, dimensioned, and/or configured to receive and house a plurality of supplies **14** of sheet material **11** therein. The dispenser housing **12** also including a discharge **15**, e.g., including one or more defined apertures or openings, that facilitates dispensing of the sheet material **11** of the supplies of sheet material **14** from the dispenser assembly **10**.

[0046] As generally shown in FIG. **1**, each supply **14** of sheet material typically includes a roll or spindle **14**A with sheet material **11** wrapped or spun thereabout. The dispenser assembly **10** further includes a plurality of support assemblies **16** rotatable supporting plurality of supplies **14** within the dispenser housing **12**. That is, each supply of sheet material **14** is configured to be supported by a corresponding support assembly **16** positioned with the chamber(s) **13** of the dispenser housing **12**. The plurality of supplies **14** of sheet material can include a first supply **18** of sheet material that is supported by a corresponding first support assembly **20**, and a second supply **22** of sheet material that is supported by a second support assembly **24**. The first and second support assemblies **20**/**24** can be spaced apart from each other along the dispenser housing **12** as generally indicated in FIG.

1. A partition or other suitable portion **25** further can be positioned between the first and second support assemblies **20**/**24**.

[0047] In one construction, the support assemblies **20**/**24** can include slots or grooves **30**/**32** defined in or along the dispenser housing **12** (e.g., in the cover **12**A and/or backing portion **12**B or other walls, portions, supports, etc. within the dispenser housing **12**). The slots **30**/**32** can be configured to at least partially receive first and second ends **34**/**36** of the support roll or spindle **38**/**40** for the first and second supplies **18**/**22** of sheet material, and at least a portion of each of the supplies of sheet material being supported by and/or resting on or engaging a corresponding guide roller **42**/**44**. The slots or grooves of the roll support assemblies **20**/**24** can include one or more angled or sloped portions **46**/**48** having a variable slope or angle to increase and/or decrease an amount of force the supply **18**/**22** of rolled sheet material exerts on the guide rollers **42**/**44**. The slope of portions **46**/**48** can be selected such that as the sheet material is fed from the supplies **18**/**22** of sheet material and is depleted (e.g., the amount and thus the weight of sheet material remaining on a roll **38**/**40** decreases), the position of the supply rolls **18**/**22** will change so as to generally maintain a substantially constant downward force exerted by the sheet material supplies **18**/**22** on the respective guide rollers **42**/**44**.

[0048] As generally shown in FIG. **2**A, the guide rollers **42**/**44** of the dispenser assembly **10** will be positioned along or substantially proximate, adjacent, etc. and engaging the supplies **14** of sheet material, with the first guide roller **42** engaging the first supply **18** of sheet material and the second guide roller **44** engaging the second supply **22** of sheet material. Each of the guide rollers **42**/**44** can include an elongated body **43**/**45** defining a substantially cylindrical sidewall **43**A/**45**A configured to engage the sheet material from the supplies **18**/**22** of sheet material, e.g., to at least partially support the supplies **18**/**22** of sheet material within the slots **30**/**32** (FIG. **1**) and to facilitate dispensing of the supplies **18**/**22** of sheet material from the dispenser assembly **10**. The body **43**/**45** of the guide rollers **42**/**44** can be formed from a plastic material, though other materials, such as wood, elastomeric materials, such as rubber, or other composite or synthetic materials or combinations thereof, can be used without departing from the scope of the present disclosure. The guide rollers **42**/**44** also can include bands **43**B/**45**B of a gripping material, e.g., including a rubber or other elastomers or synthetic materials, to assist in gripping or engaging the sheet material **11** without causing damage thereto. The guide rollers **42**/**44** are rotatably mounted within the dispenser housing **12**. FIG. **2**A shows that the guide rollers **42**/**44** can include bearing assemblies **47**/**49** attached to the guide rollers **42**/**44** that support the guide rollers **42**/**44** within the dispenser housing **12**, such that the guide rollers **42**/**44** are rotatable thereabout (e.g., the bearing assemblies **47**/**49** can be fixedly connected to the backing portion **14**B and/or the cover **14**A or other walls, portions, supports, etc. of the dispenser assembly **12**). The bearing assemblies **47**/**49** can include, for example and without limitation, roller bearings, ball bearings, and the like, or other suitable mechanisms that facilitate rotation of the guide rollers **42**/**44**.

[0049] FIGS. **1** and **2**A further show that the dispenser assembly **10** includes a dispensing system or mechanism **50** for selectively dispensing predetermined amounts (i.e., particular, selected lengths) of sheet material **11** from the plurality of supplies **18**/**22** of sheet material. The dispensing system **50** can include a plurality of driven rollers **56**/**58** for engaging and driving the sheet material from the supplies **18**/**22** of sheet material. For example, the first supply **18** of sheet material can be dispensed by a corresponding first driven roller **56** and the second supply **22** of rolled sheet material can be dispensed by a corresponding second driven roller **58**. The first driven roller **56** will engage and draw or urge sheet material from the first supply **18** of sheet material along a first discharge path **65**A toward and out of the discharge **15** of the dispenser housing **12**, while the second driven roller **58** will engage and draw or urge sheet material from the second supply **22** of sheet material along a second discharge path **65**B toward and out of the discharge of the dispenser housing **12**.

[0050] As additionally indicated in FIGS. **1** and **2**A, the dispenser assembly **10** includes a drive mechanism **60** operatively connected or coupled to the plurality of driven rollers **56**/**58** to drive rotation thereof. In one variation, the drive mechanism **60** can include a motor **60**A (e.g., a brushless servo or stepper motor, or other, similar type of variable speed, reversible electric motor), though or other suitable drive mechanisms, drive systems, actuators, and/or the like can be used without departing from the scope of the present disclosure. The driven rollers **56**/**58** positioned substantially adjacent and along the guide rollers **42**/**44** rotate under the power of the drive mechanism **60** to pull the sheet material **11** from the respective supplies **18**/**22** and along the discharge paths **65**A/B at least partially defined between the driven rollers **56**/**58** and associated guide rollers **42**/**44** and through the discharge **15** defined in the dispenser housing **12**. Each driven roller **56**/**58** further is selectively driven/rotated by a drive mechanism **60** linked to or otherwise in communication with the driven rollers **56**/**58**. The drive mechanism **60** communicates with a control circuitry **5** (e.g., including controller **100** as shown in FIG. **5**) of the dispenser assembly **10** to receive instructions and power for selectively activating and driving the driven rollers **56**/**58** of each roller assembly through a dispensing cycle (e.g., a determined time, number of revolutions, etc.), to feed the selected or desired amount/length of the sheet material through the discharge **15** of the dispenser housing **12**. In addition, the drive mechanism **60** can be driven in a first direction, e.g., D**1** in FIG. **1**, to drive the first driven roller **56** and move the sheet material from the corresponding first supply **18** of sheet material along the first discharge path **65**A toward and out from the discharge **15** of the dispenser housing **12**. The drive mechanism **60** also can be reversed and driven in a second direction, e.g., D**2** in FIG. **1**, to drive the second driven roller **58** and move the sheet material from the corresponding second supply **22** of sheet material along the second discharge path **65**B toward and out from the discharge **15** of the dispenser housing **12**.

[0051] FIG. **2**A shows that the driven rollers **56**/**58** can include an elongated body **57**/**59** with a generally cylindrical sidewall **57**A/**59**A that is configured to engage and pull the sheet material **11** from the respective supplies of sheet material **18**/**22**. The driven rollers **56**/**58** are rotatably mounted within the dispenser housing **12** by one or more bearing assemblies **61**/**63** (e.g., including roller bearings, ball bearings, etc. or other suitable bearing mechanisms that facilitate rotation of the driven rollers **56**/**58**) connected to the backing portion **12**B and/or the cover **12**A or other suitable wall, portion, support, etc. within the dispenser housing **12**. The driven rollers **56**/**58** further can include bands of a gripping material **57**B/**59**B, such as a rubber or synthetic material, to assist in pulling the sheet material between the driven rollers **56**/**58** and guide rollers **42**/**44**, without causing damage to the sheet material as it passes between the driven **56**/**58** and guide rollers **42**/**44**.

[0052] In some constructions, the driven rollers **56**/**58** and/or the guide roller **42**/**44** can be biased into engagement with each other (e.g., by one or more biasing mechanism, such as springs, e.g., compression springs, tension springs, torsion springs, and the like; elastic cylinders; and/or other suitable biasing mechanisms) to press or otherwise engage the sheet material between the driven rollers **56**/**58** and guide rollers **42**/**44**. The roller assemblies **52**/**54** further can include additional guide or pressing rollers positioned adjacent the driven rollers **56**/**58** and/or guide rollers **42**/**44** to guide and/or engage the sheet material without departing from the scope of the present disclosure.

[0053] In addition, the drive system **50** can include a belt driven transmission assembly **62** including a driven belt **62**A operatively connecting or engaging the driven mechanism **60** and driven rollers **56**/**58** to transfer power therebetween for selectively driving rotation of the first driven roller **42** and/or the second driven roller **44**. For example, as indicated in FIGS. **1** and **2**A, the drive mechanism **60** can be operatively connected to each of the driven rollers **56**/**58** by a drive belt **62**A that engages corresponding belt pulleys or belt gears **67**/**69** connected to each of the driven rollers **56**/**58** and a belt pulley or belt gear **71** connected to the driven mechanism **60**. The belt gears **67**, **69**, and **71** can include a first driven roller belt gear **67** operatively connected to the first driven roller **42**; a second driven roller belt gear **69** operatively connected to the second driven roller **44**; and a drive mechanism belt gear **71** operatively connected to the drive mechanism **60**.

[0054] In the illustrated construction, a single belt **62** is shown operatively connected to the drive mechanism **60** (e.g., engaging the belt gear **71** that is coupled to a driveshaft **60**B of the motor **60**A) and to each of the driven rollers **56**/**58** (e.g., engaging the belt gears **67**/**69** attached thereto or otherwise in operative communication therewith); however, it is contemplated that a series of belts can be used to connect the drive mechanism **60** and driven roller **56**/**58**, such as one drive belt connecting the drive mechanism **60** and driven roller **56** and another drive belt connecting the drive mechanism **60** and driven roller **58**, without departing from the scope of the present disclosure. It further will be understood that in additional or alternative constructions, one or more of the driven rollers **56**/**58** can be connected to the driven mechanism **60** by other suitable transmission assemblies or mechanisms, such as a series of gears or other suitable transmission assemblies.

[0055] In an additional or an alternative construction, as generally indicated in FIG. **2**C, the belt gears **67**/**69** can be operatively connected to the rollers **42**/**44** (rather than rollers **56**/**59**) such that the rollers **42**/**44** are driven rollers. That is, as FIG. **2**C indicates, the belt **62**A can engage the belt gears **67**/**69** attached to the ends of the driven rollers **42**/**44** such that the driven rollers **42**/**44** can be selectively driven and rotated by the drive mechanism **60**. In this construction, the rollers **56**/**58**, which are not directly engaged by the belt **62**A, are allowed to float, and further can be biased into engagement with the driven rolls **42**/**44** (e.g., by one or more biasing assemblies including at least one biasing member, such as a spring, biased cylinder, etc.). The rollers **56**/**58** accordingly can be configured as guide or pressing rollers to help to direct the sheet material along the respective discharge paths **65**A and **65**B. In additional variations, at least one or a plurality of pressing or guide rollers can be positioned along and biased into engagement with the driven rollers **42**/**44**. Still further, the pressing or guide roller(s) (e.g., rollers **56**/**58**) can be coupled to the rollers **42**/**44** by a transmission mechanism, such as a belt driven transmission mechanism, that can transfer power between the rollers **42**/**44** and **56**/**58** and also can be configured to bias the rollers **42**/**44** and **56**/**58** towards engagement with one another.

[0056] As shown in FIGS. **1**-**3**B, the belt **62**A also can include a plurality of cogs or teeth **62**B disposed thereabout and configured to engage corresponding notches, teeth, and the like in the belt gears, i.e., **67**, **69**, and/or **71**. The belt **62**A and/or the cogs **62**B thereof can be formed from a rubber material, such as a chloroprene rubber, or other suitable rubber, though any suitable material can be used without departing from the scope of the present disclosure. The belt **62**A also can include one or more layers or plies, including a tensile layer that comprises a reinforcement, for example, fiberglass, though the belt can comprise any suitable material, e.g., other rubbers, plastics, synthetics and/or composites, without departing from the present disclosure. Additionally, the belt **62**A can include a wrapping, such as a cloth or sheet material comprising high elastic nylon, though the wrap cloth can comprise any other suitable material without departing from the present disclosure.

[0057] The driven rollers **56**/**58** (or driven rollers **42**/**44** as shown in FIG. **2**C) generally are configured to be selectively rotatable to dispense amounts of sheet material **11** from their corresponding supply of sheet material **18** or **22** when driven in one direction by the drive mechanism **60**, but generally will remain substantially stationary, such that sheet material **11** is not dispensed from its corresponding supply of sheet material **18** or **22**, when the drive mechanism **60** is driven in the opposite direction. For example, when the first driven roller **56** is rotated by the drive mechanism **60** in a first direction D**1** shown in FIG. **1**, the first driven roller **56** can engage and feed/dispense sheet material from the first supply **18** of sheet material, while the second driven roller **58** remains generally stationary such that sheet material from the second supply **22** is not dispensed therefrom. When the drive mechanism **60** is driven in a second, opposite direction D**2** shown in FIG. **1**, the second drive roller **58** will be rotated to dispense the select/predetermined amounts of sheet material from the second supply **22** of sheet material while the first driven roller **56** remains generally stationary, such that the sheet material is not dispensed from the first supply **18** of rolled sheet material. Accordingly, the dispenser assembly **10** can provide for selective dispensing of the plurality of supplies **18** or **22** of sheet material by controlling the driving direction of the drive mechanism **60**. Thus, sheet material **11** can be dispensed from one supply of sheet material **18** or **22**, until such supply is substantially dispensed or exhausted, after which the direction of the drive mechanism **60** can be switched/changed (e.g., reversed or otherwise altered) to transfer to and begin dispensing the sheet material **11** from the other supply of sheet material **18** or **22**.

[0058] The driven rollers **56**/**58** (or driven rollers **42**/**44** as shown in FIG. **2**C) also can include or incorporate a clutch assembly or mechanism **70**, such as a hybrid or one-way clutch mechanism, that allows for selective transfer of power between the drive mechanism **60** and the driven rollers **56**/**58** (or driven rollers **42**/**44** as shown in FIG. **2**C), such as generally shown in FIGS. **2**A and **2**B. For example, as FIGS. **2**A-**2**C indicate, the clutch assembly **70** can be incorporated or integrated with the belt gears **67**/**69** connected to the driven rollers **56**/**58** (or rollers **42**/**44** as shown in FIG. **2**C). Accordingly, when the drive mechanism **60** is driven in a first direction D**1**, the clutch assembly **70** of the first driven roller **56** will lock/engage for transfer of power/torque to the first driven roller **56** so that the first driven roller **56** is driven by the drive mechanism **60** and rotated to dispense its corresponding supply **18** of sheet material (while the clutch assembly **70** of the second driven roller **58** remains generally disengaged such that the second driven roller **58** is substantially stationary as no power/torque is transferred from the drive mechanism **60** and the second driven roller **58**). In addition, when the drive mechanism **60** is driven in the opposite direction D**2**, the clutch assembly **70** for the first driven roller **56** will unlock or disengage such that there is no transfer of power/torque between the drive mechanism **60** and the first driven roller **56** such that the first driven roller **56** remains generally stationary (while the clutch assembly **70** for the second driven roller **58** engages or locks for transfer of power/torque to the second driven roller **58** so that the second driven roller **58** is rotated to dispense its corresponding supply **22** of sheet material).

[0059] In one example construction, as generally indicated in FIG. **2**B, each clutch assembly **70** can include one or more tracks/races, such as inner and outer races **72**/**74**, that rotate together (when engaged) or independently of one another (when disengaged). The clutch assembly **70** further can include a plurality of biased rollers or bearings **76** can be received between the inner and outer races and can be biased such as by a series of springs **78** or other biasing mechanisms, toward/against corresponding surfaces or other engagement portions **79** of the outer race **74** to stop or prevent rotation of the bearings **76** and provide engagement or coupling between the inner **72** and outer **74** races. For example, as indicated in FIG. **2**B, when the inner race **72** is rotated in the direction D**1** shown in FIG. **2**B upon rotation of the driven mechanism **60**, the bearings **76** are engaged and urged into the surfaces **79**, which blocks or prevents rotation of the rollers **76**, allowing the inner race **72** to engage, drive, and rotate the outer race **74** and thus rotate the driven roller **58** to facilitate feeding of sheet material from its corresponding supply **22**. And, when the inner race **72** is rotated in the opposite direction D**2** shown in FIG. **2**B, the rollers **76** move away from and do not engage the outer race **74** (e.g., do not engage the engagement portions **79**) under the control of the springs **78**, such that the rollers **76** can rotate or spin freely allowing the inner race **72** to turn independently of the outer race **74**, such that the driven roller **58** does not rotate and remains generally stationary.

[0060] The dispenser assembly **10** further can include a tensioning assembly **80** including one or more biasing members **82**. For example, as shown in FIGS. **1**, **2**A, and **3**A-B, the one or more biasing members **82** can be operatively connected to the drive mechanism **60** for biasing the drive mechanism **60**, such as to provide tension along the drive belt **62**A (e.g., to substantially prevent, reduce, or inhibit wear, slippage, etc. thereof) and/or to provide dampening for the drive mechanism **60** (e.g., dampening or absorbing motor vibrations or other components of the drive system). In one example, the biasing member(s) **82** can include a tension spring(s) **82**A with one end **83** thereof operatively connected to the drive mechanism **60** (or part/component connected to the drive mechanism **60** or a bracket, support, frame, etc. supporting the drive mechanism within the dispenser housing **12**) and another end thereof **85** operatively connected to a portion of the dispenser housing **12**.

[0061] FIGS. **3**A and **3**B illustrate perspective and cross-sectional views of a tensioning assembly **80** according to one example construction of the present disclosure. As indicated in FIGS. **3**A and **3**B, the tensioning assembly **80** can include a support assembly **90** including a bracket **92** that is connected to and supports the drive mechanism **60** (i.e., the motor **60**A and the belt gear **71** attached thereto) and that is movably connected to the dispenser housing **12** (e.g., movably connected to a wall, support, etc. **94** of, or otherwise connected to, the dispenser housing **12** (FIG. **3**B). The bracket **92** further includes one or more connection mechanisms **93** that are configured to connect to the biasing member(s) **82**. That is, one hooked, or looped end **83** of the biasing member(s) **82** can be connected to the connection mechanism **93** (e.g., including a rod **93**A or other suitable connection mechanism, such as a hooked or looped connection mechanism), and the opposite, hooked or looped end **85** of the biasing member **82** can be operatively connected to a wall, support or other suitable portion **94** of the dispenser housing **12** (e.g., via a hooked or looped connection mechanism **94**A or other suitable connection mechanism, such as a rod, projecting portion, etc.). Accordingly, the tensioning assembly **80** provide tension, e.g., a tensile force or stresses, along the drive belt **62**A (e.g., to substantially prevent, reduce, or inhibit slippage, premature wear, etc. thereof) and also to provide dampening for the dispenser assembly **10** during operation thereof (e.g., to dampen or absorb vibrations of the motor **60**A, or other components of the drive assembly, such as to reduce noise generated thereby).

[0062] The bracket **92** can include a first portion or section **96** that is connected to the motor **60**A, and a second portion or section **98** that is movably connected to the wall **94** of the dispenser housing **12**. The first portion **96** of the bracket **92** can be connected to the motor **60**A by one or more fasteners **100**, such as screws, bolts, and the like. For example, the fasteners **100** can be received through holes **102** (e.g., threaded or unthreaded holes) defined through the first portion **96** and can also be tightened into or otherwise received in corresponding threaded holes **104** of the motor **60**A to secure the motor **60**A to the first portion **96**. The first portion **96** further can include a flange or projecting portion **96**A that defined a passage or opening **96**B that is sized, dimensioned, and/or configured for receipt of the motor **60**A, e.g., to facilitate a frictional or snap fitting between the motor **60**A and the first portion **96**.

[0063] The first portion **96** further can be connected to the second portion **98** by support rods or posts **106**, one or more of which can be integrally formed with the first **96** and/or second **98** portions, as generally shown in FIGS. **3**A and **3**B. The support rods **106** further include a passage or opening defined therethrough, which can include threads or be unthreaded and allow for the receipt of a fastener, such as for example and without limitation, a bolt, screw, and/or the like, that can be received through corresponding holes in the first **96** and/or second **98** portions to facilitate attachment of the first **96** and/or second **98** portions. The support rods **106** can be otherwise attached to the first **96** and/or second **98** portions, such as, for example and without limitation, using an adhesive, frictional or fitted connection, without departing from the scope of the present disclosure.

[0064] As additionally indicated in FIGS. **3**A and **3**B, the tensioning assembly **80** can include a movable connection mechanism **110** that movably connects the second portion **98** to a wall **94** of the dispenser housing **12**, i.e., such that the bracket **92** can move under the guidance or control of the biasing member(s) **82**. In one construction, the moveable connection mechanism **110** can include a bearing assembly **112** that is rotatably or pivotally connected to the wall **94** of the dispenser housing **12**. The bearing assembly **112** can include one or more roller bearings or other suitable bearings, bushings, or mechanisms that allow for pivoting or rotation of the bracket about the bearing assembly **112**. In an alternative construction, the connection mechanism **110** can include a plurality of fasteners, such as, for example and without limitation, screws, bolts, and the like, and the second portion **98** of the bracket **92** can be connected to the wall **94** by the plurality of fasteners, which can be received within slots or other elongated apertures defined in the wall **94** to allow for sliding movement of the bracket **92** under the guidance or control of the biasing member(s) **82**.

[0065] FIGS. **3**A and **3**B further show that the second portion **98** of the bracket **92** can at least partially support the belt gear **71** connected to the driveshaft **60**B of the motor **60**A, as well as the driveshaft **60**B, itself. For example, the tensioning assembly **80** can include a belt gear bearing assembly **120** (for example and without limitation, ball bearings, roller bearings, and/or the like) that is at least partially received within and engages an opening or passage **122** defined within a flange or projecting portion **124** of the second portion **98** of the bracket **92** (e.g., such that the bearing assembly **120** is supported thereby), and that also engages the belt gear **71**. For example, the bearing assembly **120** engages a flange or other projecting portion **126** formed with the belt gear **71** (e.g., the flange **126** is at least partially fitted into or otherwise received within a passage **128** of the bearing assembly **120**). Accordingly, the bracket **92** at least partially supports the belt gear **71** and/or driveshaft **60**B of the motor **60**A, e.g., such that the motor **60**A and belt gear **71** move as a substantially unitary structure to help to reduce, inhibit, or prevent bending, twisting, or other unwanted movement of the driveshaft **60**A and/or belt gear **71** due to the urging of the biasing member **82** and/or operation of the dispenser assembly **10**. This further can help to reduce or inhibit premature and/or uneven wear or other damage to the motor **60**A, belt gear **71**, and/or other components of the drive assembly or dispenser assembly.

[0066] The dispenser assembly **10** also can include a cutting mechanism/assembly **150** for cutting or severance of dispensed sheet material. In one construction, as shown in FIGS. **1**, **4**A, and **4**B, the dispenser housing may include one or more tear bars or other suitable cutting members **151** disposed adjacent or along the discharge **15** of the dispenser housing **12** so that a user can separate a sheet or measured amount of the material by grasping and pulling the sheet across the tear bar **151**. In addition, or in alternative constructions, the dispenser assembly **10** can include one or more cutting mechanisms that are incorporated with the guide rollers **42**/**44** and/or the driven rollers **56**/**58** and are configured to move with rotation thereof to cut, sever, and/or perforated the sheet material **11** as or after it is dispensed from the supplies **18** or **22** of sheet material.

[0067] As additionally shown in FIGS. **1**, **4**A, and **4**B, the dispenser assembly **10** can include a pawl member assembly **149** including a pivotally mounted pawl member **152** that is located proximate to the tear bar **151** such that movement of sheet material into the tear bar **151** for severance pivots the pawl member **152** between multiple positions **152**A/**152**B. The pawl member assembly **149** also includes a signal device **153**, such as a proximity sensor switch or the like, cooperative with the pawl member **152**, that is arranged such that movement of the pawl member **152** between various positions causes the signal device **153** to send a signal to notify the control circuit or controller **5** that the sheet material has been removed. That is, movement of the sheet material into the cutting mechanism **150** generally will move the pawl member **152** from a first position **152**A to a second position **152**B, which activates the signal device to transmit one or more signals to the control circuitry **5** to notify the control circuit **5** that a portion of the dispensed sheet material has been removed. By way of example, such signal device **153** responsive or cooperative with the pawl member **152** can include an infrared emitter and detector that detects movement of the pawl member **152** between first **152**A and second **152**B positions, though any suitable sensor or detection mechanism can be employed such as, for example and without limitation, a proximity sensor or other detector, a magnetic switch, a mechanical switch, or the like.

[0068] After receiving a signal that sheet material may have been removed, the control circuitry **5** further can activate a sheet material detection sensor **158** (FIGS. **1** and **5**) to verify that the sheet material has been removed from the discharge **15**. The sheet material detection sensor **158** can include an emitter **158**A/B and a detector **158**A/B on opposing sides of and focused across at least a portion of one or more of the discharge paths **65**A/B. One or more signals transmitted from the sheet material detection sensor **158** can indicate that sheet material is present or absent from the discharge path **65**A/B or discharge **15** (e.g., indicating that sheet material has been removed by a user). The sheet material detection sensor **158** further can be activated by the control circuitry **5** of the dispenser assembly **10** to verify that sheet material has been removed from the discharge **15**. Examples of pawl members and sheet material detection sensors are shown and described in U.S. patent application Ser. No. 13/155,528, the disclosure of which is incorporated herein by reference as if set forth in its entirety.

[0069] The control circuitry **5** can change the driving direction of the driving mechanism **60** based on signals received from the pawl member assembly **149** and/or the sheet material detection sensor **158**, e.g., to reverse the motor **60**A and alternate dispensing between the supplies **18**/**22** of sheet material. For example, if the control circuitry **5** receives one or more signals from the signal detection device **153** and/or the sheet material detection sensor **158** that indicate that sheet material cannot be dispensed from one of the supplies **18** or **22** of sheet material (e.g., indicating an error condition, sheet material jam, etc. or that the sheet material has been exhausted from the supply **18** or **22**), the control circuitry **5** can generate and transmit one or more signals to the drive mechanism **60** to change the driving direction thereof to dispense from the other supply **18** or **22** of sheet material. In addition, signals received from the signal device **153** and/or the sheet material detection sensor **158** can be used by the control circuitry **5** to calculate, estimate, or otherwise determine a supply level or amount of sheet material remain in the supplies **18** or **22** of sheet material. In one example, the control circuitry **5** can determine the supply level based on the number of times signals are received from the signal device **153** and/or the sheet material detection sensor **158** (e.g., the original amount of sheet material, the lengths of sheet material being dispensed, and the number of activation times for the pawl member **152** and/or sheet material detection sensor **158** can be used to determine the remaining amount of sheet material in the supply). And, when the supply level is at or below a threshold level, such as, for example and without limitation, exemplified threshold levels of 0%, 5%, 15%, and the like, the control circuitry **5** can generate one or more signals to change the direction of the motor **60**A and dispense the sheet material from the other supply. The control circuitry **5** further can generate and transmit one or more alerts, alarms, notifications, if/when the control circuitry **5** determines that one or both of the supplies **18**/**22** are below a threshold level, such as, for example and without limitation, exemplified threshold levels 0%, 5%, 15%, 30%, and the like, and/or one or more signals received from the signal device **153** and/or the sheet material detection sensor **158** indicate an error condition, sheet material jam, etc.

[0070] The dispenser assembly **10** further can include a monitoring system **200** in communication with the control circuitry **5** (e.g., with the controller **100** thereof as shown in FIG. **5**) and configured to determine a supply level or remaining amount of sheet material of the supplies **18**/**22** of sheet material. In response to such information/determination, the control circuitry **5** can initiate or change the direction of the motor, e.g., when an amount of remaining sheet material is sensed less than a threshold volume. In one construction, as generally indicated in FIG. **1**, the monitoring system **200** can include magnets **202** connected to the support rolls **38**/**40** of the first and second supplies **18**/**22** of sheet material supply, with the magnets **202** being rotatable therewith during dispensing thereof. In one construction, as indicated in FIG. **1**, the monitoring system **200** can include a single magnet **202** connected to the support rolls **38**/**40**; however, a plurality of magnets, for example and without limitation, a ring of magnets with alternating polarities, can be arranged along the support rolls **38**/**40**, without departing from the scope of the present disclosure. In addition, or in alternative constructions, the monitoring system **200** can include a magnet **202** and/or magnets connected to the guide rollers **42**/**44** (FIG. **2**A) and/or the driven rollers **56**/**58** (FIG. **2**B).

[0071] In addition, as shown in FIGS. **1** and **2**A-**2**B, the monitoring system **200** can include a sensor **204** arranged substantially proximal or adjacent each magnet **202** or plurality of magnets. The sensor **204** can include, for example and without limitation, a reed switch, a hall element, proximity sensor, or other suitable sensor operable to measure or otherwise capture variations, fluctuations or other changes in a magnetic field generated as each corresponding magnet **202**, or plurality of magnets, is rotated with the supplies **18**/**22** of sheet material, guide rollers **42**/**44**, and/or driven rollers **56**/**58** during dispensing and passes by the corresponding sensor **202**. The detected variations, fluctuations or changes of the magnetic field can be correlated to number of rotations of the supplies of sheet material **18**/**22**, guide rollers **42**/**44**, and/or driven rollers **56**/**58**, and/or a rotation angle of the supplies of sheet material **18**/**22**, guide rollers **42**/**44**, and/or driven rollers **56**/**58** for dispensing a desired length of the sheet material during each dispensing operation. By substantially continuously monitoring the number of rotations of the supplies of sheet material **18**/**22**, guide rollers **42**/**44**, driven rollers **56**/**58**, and/or the number of rotations the driving mechanism **60** during dispensing operations, a diameter of the supplies **18**/**22** of sheet material can be substantially dynamically or continuously determined during or following each dispensing operation (e.g., the diameters can be determined during or after each dispensing operation) and, based on this determined/monitored diameter, an amount of sheet material remaining likewise can be dynamically determined, e.g., by the controller **100** of the control circuitry **5** based on signals received from the monitoring system **200**. Additionally, other sensing devices or mechanisms, such as encoders or other detectors that can monitor and provide a measurement of the number of rotations of the supplies of sheet material **18**/**22**, guide rollers **42**/**44**, driven rollers **56**/**58**, and/or drive mechanism **60** can be used, without departing from the scope of the present disclosure. One example monitoring system is described in U.S. patent application Ser. No. 15/922,157 which is incorporated by reference herein as if set forth in its entirety.

[0072] Furthermore, when the processor **100** of the control circuitry **5** determines that the supply level of one of the supplies **18** or **22** is at or below a threshold level, such as, for example and without limitation, exemplified threshold levels of 0%, 5%, 15%, and the like, based on one or more signals received from the monitoring system **200**, the control circuitry **5** can generate one or more signals to change the direction of the motor **60**A and dispense the sheet material from the other supply **18** or **22**. In particular, upon a determination that the supply level of the first supply **18** of sheet material is below a threshold level, the direction of the drive mechanism can be changed from the first direction D**1** in FIG. **1** to the second direction D**2** in FIG. **1** to dispense the sheet material **11** from the second supply **22** of sheet material. Likewise, upon a determination that the supply level of the second supply **22** of sheet material is below a threshold level, the direction of the drive mechanism **60** can be changed from the second direction D**2** in FIG. **1** to the first direction D**1** in FIG. **1** to dispense the sheet material **11** from the first supply **22** of sheet material. The control circuit **5** further can generate and transmit one or more alerts, alarms, notifications, if/when the control circuit **5** determines that the supply level of one or both of the supplies **18**/**22** is below a threshold level,, such as, for example and without limitation, exemplified threshold levels of 0%, 5%, 15%, 30%, 40%, and the like.

[0073] In addition, or in the alternative, a switch **210** disposed along the dispenser housing **12** can be manually activated by a system operator to change the direction of the dispensing mechanism **60**, e.g., between directions D**1** and D**2** shown in FIG. **1**; though the direction can be changed using any suitable means, such as, for example and without limitation, an electronic device (e.g., computer, smart phone, tablet, and the like) configured to be managed by a system operator can be used to change the direction of the drive mechanism **60**. For example, the control circuitry **5** can include one or more receivers/transmitters configured to communication with the electronic device, and the control circuitry **5** can change the direction of the drive mechanism based on one or more signals received from the electronic device.

[0074] FIG. **5** illustrates a block diagram of the electronic control system or control circuitry **5** for operating the dispenser assembly **10** in an exemplary embodiment. The control circuitry generally includes a controller **100** that can include one or more processors (e.g., microprocessors) and one or more memories (e.g., RAM, ROM, etc.). One or more of the memories can store instructions, workflows, control software, and the like that are accessed and executed by the processor for carrying out operations or functions of the dispenser assembly **10**. The dispenser or operative components of the dispenser may be powered by a power supply **154** such as one or more batteries **155** contained in a battery compartment of the dispenser housing **12**, though any suitable battery storage device may be used for this purpose. Alternatively, or in addition to battery power, the dispenser may also be powered by a building's power distribution system, such as the exemplified alternating current (AC) distribution system as indicated at **156**. For this purpose, a plug-in modular transformer/adapter can be provided with the dispenser assembly **12**, which connects to a terminal or power jack port located, for example, in the bottom edge of the circuit housing for delivering power to the control circuitry and associated components. The control circuitry **5** also may include a switch, such as, for example and without limitation, a mechanical or electrical switch, that can be configured to isolate the battery circuit upon connecting the AC adapter in order to protect and preserve the batteries.

[0075] In one example, a sensor, such as a proximity detector or other sensor **160**, may be configured to detect an object placed in a detection zone external to the dispenser assembly **10** to initiate operation thereof. This sensor **160** may be a passive sensor that detects changes in ambient conditions, such as, for example and without limitation, ambient light, capacitance changes caused by an object in a detection zone, and the like. In an alternate embodiment, the sensor **160** may be an active device and include an active transmitter and associated receiver, such as, for example and without limitation, one or more infrared (IR) transmitters and an IR receiver. The transmitter transmits an active signal in a transmission cone corresponding to the detection zone, and the receiver detects a threshold amount of the active signal reflected from an object placed into the detection zone. The control circuitry **5** generally will be configured to be responsive to the sensor for initiating a dispense cycle upon a valid detection signal from the receiver. For example, the proximity sensor **160** or other detector can be used to detect the presence of a user's hand. In some variations, the sheet material detector sensor **158** also can be aligned to detect a user's hand below the dispenser assembly **10** and can include a second infrared emitter/detector pair aligned to detect a sheet hanging in or below the discharge **15**.

[0076] The controller **100** of the control circuitry can control activation of the dispensing mechanism upon valid detection of a user's hand for dispensing a measured length of the sheet material. In one variation, the control circuitry **5** can track the running time of the motor **60**A, and/or receive feedback information directly therefrom indicative of a number of revolutions of the driven roller and correspondingly, an amount of the sheet material feed thereby. In addition, or as a further alternative, as discussed, monitoring systems, such as, for example and without limitation, sensors, and the like, and associated circuitry may be provided for this purpose. Various types of sensors can include, for example and without limitation, IR, radio frequency (RF), capacitive or other suitable sensors, and any one or a combination of such sensing systems can be used. The control circuitry **5** also can control the length of sheet material dispensed. Any number of optical or mechanical devices may be used in this regard, such as, for example and without limitation, an optical encoder may be used to count the revolutions of the guide or driven rollers, with this count being used by the control circuitry **5** to meter the desired length of the sheet material to be dispensed.

[0077] The processing logic for operation of the dispenser assembly **100** in, for example, and without limitation, hand sensor and butler modes, can be part of the control software stored in the memory of the controller **100** of the control system **5**. One or more binary flags can also be stored in memory and represent an operational state of the dispenser (e.g., “sheet material cut” set or cleared). An operational mode switch in dispenser can be configured to set the mode of operation. For example, in the hand sensor mode, the proximity (or hand) sensor **160** can detect the presence of a user's hand below the dispenser housing **12** and, in response, the drive mechanism **60** can be actuated or otherwise operated to dispense a measured amount of sheet material from one of the supplies **18** or **22**. Subsequently, the control circuitry **5** can be configured to monitor when the sheet of material is removed. For example, actuation of the pawl member **152** or triggering/activation of a sheet material detection sensor **158** can determine the removal of sheet material and reset the proximity sensor **160**. The proximity sensor **160** also can configured to control or otherwise not allow additional sheet material to be dispensed until the proximity sensor is reset. It is contemplated that, if the proximity sensor **160** detects the presence of a user's hand but does not dispense sheet material, the control circuit can be configured to check for sheet material using the sheet material detection sensor **158**. If sheet material has not been dispensed (i.e., no sheet material is hanging from the dispenser), the drive mechanism **60** can be activated to dispense a next sheet.

[0078] A multi-position switch **162** also can be provided to switch the dispenser operation between a first or standard operation mode and a second mode, such as a butler mode. In the butler mode, the proximity sensor **160** that is configured to detect the presence of a user's hand/object can be deactivated, and the controller **100** can automatically dispense sheet material when the cover is closed and the dispenser assembly **10** is put into operation. The sheet material detection sensor **158** further can determine if a sheet is hanging from the dispenser. If sheet material is hanging, the controller **100** will then monitor when the sheet of material is removed. For example, a cutting mechanism movement detector, which can be configured to detect actuation or movement of the cutting mechanism; the pawl member **152**; and/or the sheet material detection sensor **158** can determine the removal of sheet material and can subsequently reset the dispenser assembly **10**. The next sheet will be dispensed automatically. However, if the sheet material detection sensor **158** determines the absence of hanging sheet material, the drive mechanism **60** will be activated to dispense the next sheet. Subsequently, the controller **100** will determine if the sheet has been removed before dispensing another sheet.

[0079] Optionally, the dispenser assembly **10** is configured to be operative in the first mode and, as such, to be responsive to a signal from the proximity sensor **160** to dispense a sheet of material. Optionally, the dispenser assembly **10** is operative in the second mode to dispense a next sheet in response to the signal means being activated by movement of the pawl member **152** in response to dispensed sheet material being removed from the dispenser assembly **10**. In another optional variation, the dispenser assembly **10** can be configured to operate in a second mode to dispense a next sheet in response to the signal means **153** being activated by movement of the pawl member **152** and a signal from a sheet material detection sensor **158** that the sheet material has been removed from the dispenser assembly **10**.

[0080] The dispenser assembly **10** generally can be configured to dispense a measured length of the sheet material, which may be accomplished by various means, such as, for example and without limitation, a timing circuit that actuates and stops the operation of the motor **60**A driving the driven rollers **56**/**58** after a predetermined time. In one optional variation, the motor **60**A can be configured to provide direct feedback as to the number of revolutions of the driven rollers **56**/**58**, which is indicative of an amount of the sheet material fed thereby. Alternatively, a motor revolution counter can be configured to measure the degree of rotation of the driven rollers **56**/**58** and is operatively interfaced with control circuitry **5** (e.g., the controller **100** thereof) to stop the motor **60**A after a defined number of revolutions of the motor **60**A and/or the driven rollers **56**/**58**. This motor revolution counter may be, for example and without limitation, an optical encoder type of device, a mechanical device, or the like. The control circuitry **5** can optionally include a device to allow maintenance personnel to adjust the sheet length by increasing or decreasing the revolution counter set point. In a further aspect, the multi-position switch **162** can also be configured to be in operable communication with the control circuitry **5** to select one of a plurality of time periods as a delay between delivery of an initial sheet and delivery of a next sheet to the user. Embodiments of the present disclosure described herein can also utilize concepts disclosed in U.S. Pat. Nos. 7,213,782 and 7,370,824, both of which are incorporated by reference herein as if set forth in their entireties, as well as U.S. patent application Ser. No. 13/155,528, which also is incorporated by reference herein as if set forth in its entirety.

[0081] FIGS. **6-18** show a dispenser assembly **410** for dispensing a rolled sheet material **11** and a dispensing system or mechanism **450** for a dispenser according to another embodiment of the disclosure, which embodiment is generally similar to the prior embodiments except for variations noted and variations that will be apparent to one of ordinary skill in the art. Accordingly, similar or identical features of the embodiments have been given like or similar reference numbers. In the embodiment of FIGS. **6-18**, the dispenser assembly **410** is similar to the dispenser assembly **10** of FIGS. **1-5** except that the dispensing system **450** includes a gear driven transmission assembly **462** (FIGS. **10** and **14-18**) rather than the belt driven transmission assembly **62** of the dispensing system **50** (FIGS. **1-2**C).
[0082] As shown in FIGS. **6-8**, the dispenser assembly **410** can include the dispenser housing **12** having the cover **12**A that is movable/removable (e.g., FIG. **8**) to allow access to the components of the dispenser assembly **410**, and the backing portion **12**B that is configured to mount or otherwise connect via, for example and without limitation, fasteners, adhesive, and the like, to the dispenser assembly **410** to a wall, partition, or other suitable support within a facility, such as a restroom, hospital room, and the like. The dispenser housing **12** further includes the one or more chambers or compartments **13** defined therein and sized, dimensioned, and/or configured to receive and house the plurality of supplies **14** of sheet material **11** therein (e.g., the first supply **18** and the second supply **22** of sheet material). The dispenser housing **12** also including a discharge **15** that facilitates dispensing of the sheet material **11** of the supplies of sheet material **14** from the dispenser assembly **410**.
[0083] As shown in FIGS. **8-11**, the dispensing system **450** can include driven rollers **56**/**58** for engaging and driving the sheet material **11** from the supplies **18**/**22** of sheet material. For example, the first supply **18** of sheet material can be dispensed by the corresponding first driven roller **56** and the second supply **22** of rolled sheet material can be dispensed by the corresponding second driven roller **58** similar or identical to prior embodiments. In optional embodiments, the driven rollers **56**/**58** can be mounted between a first end plate **451**A and a second end plate **451**B of the dispensing system **450** such as by exemplified bearing assemblies or other suitable features. In other aspects, the dispensing system **450** can include guide rollers **442**A/**444**A (FIGS. **8-12**) extending along the respective driven rollers **56**/**58** and guide rollers **442**B/**444**B (FIG. **12**) extending along the respective driven rollers **56**/**58**.
[0084] As exemplarily shown in FIGS. **8**, **10**, and **13-18**, the guide rollers **442**A/**444**A and **442**B/**444**B can be mounted to the end plates **451**A/**451**B in respective slots **453** (e.g., FIGS. **13** and **14**) such as by the exemplified bearing assemblies or other suitable features mounted in the slots **453**. In additional embodiments, the bearings can be movable along the slots **453** so that the guide rollers **442**A/**444**A and **442**B/**444**B can be operatively biased toward and/or against the respective driven rollers **56**/**58** such as, for example and without limitation, by springs or other suitable biasing members **466** that are configured to urge the bearings in the slots **453** toward the driven rollers **56**/**58**. For example and without limitation, the biasing members **466** can be mounted to the end plates **451**A/**451**B so that the biasing members press or are otherwise urged against the ends of the guide rollers (e.g., the bearing assemblies). Accordingly, the guide rollers **442**A/**444**A and **442**B/**444**B can press or otherwise engage the sheet material **11** against the respective driven rollers **56**/**58** as it passes between the driven rollers **56**/**58** and guide rollers **442**A/**444**A and **442**B/**444**B.
[0085] In this aspect, the dispensing system **450** can be configured to include any suitable number of guide or pressing rollers positioned adjacent the driven rollers **56**/**58** and/or guide rollers **442**A/**444**A and **442**B/**444**B to guide and/or engage the sheet material without departing from the scope of the present disclosure. In optional embodiments, the end plates **451**A/**451**B can be connected by horizontal supports to form a support structure, which can be a unitary/integral structure or separate pieces coupled together. As exemplarily shown in FIGS. **7-12**, curved guide plates **455** can be mounted between the end plates **451**A/**451**B for guiding the sheet material **11** to

the discharge **15**.

[0086] As additionally exemplarily indicated in FIGS. **10**, **12**, and **14-18**, the dispenser assembly **410** can include a drive mechanism **460** operatively connected or coupled to the driven rollers **56**/**58** via a gear driven transmission assembly **462** to selectively drive rotation thereof. In this aspect, the drive mechanism **460** can include a motor **60**A (FIG. **12**) or other suitable actuator and the driven rollers **56**/**58** are configured to rotate under the power of the drive mechanism **460** to pull the sheet material **11** from the respective supplies **18**/**22** and along the discharge paths at least partially defined between the driven rollers **56**/**58** and associated guide rollers **442**A/**444**A and **442**B/**444**B and through the discharge **15** defined in the dispenser housing **12**. In this aspect, the drive mechanism **460** is configured to operatively communicate with a control circuitry (e.g., including controller **100** as shown in FIG. **5**) of the dispenser assembly **410** to receive instructions and power for selectively activating and driving the driven rollers **56**/**58** of each roller assembly through a dispensing cycle (such as, for example and without limitation, a determined time, number of revolutions, and the like) to feed the selected or desired amount/length of the sheet material through the discharge **15** of the dispenser housing **12**.

[0087] As described in more detail below, the motor **60**A can be driven in a first direction, e.g., D**1** in FIG. **16**, to cause the transmission assembly **462** to engage and drive the first driven roller **56** and move the sheet material from the corresponding first supply **18** of sheet material along the first discharge path toward and out from the discharge **15** of the dispenser housing **12**. Similarly, the motor **60**A can be driven in a second direction, e.g., D**2** in FIGS. **17** and **18**, to cause the transmission assembly **462** to disengage with the first driven roller **56** and to engage and drive the second driven roller **58** and move the sheet material from the corresponding second supply **22** of sheet material along the second discharge path toward and out from the discharge **15** of the dispenser housing **12**.

[0088] In embodiments, the gear driven transmission assembly **462** can optionally also include a plurality of gears for transferring power from the drive mechanism **460** to a selective one of the driven rollers **56**/**58**. In this aspect and as shown in FIGS. **10** and **14-18**, the transmission assembly **462** can include a drive gear **471** mounted to the motor **60**A via a drive shaft extending through an opening in the first end plate **451**A of the dispensing system **450**. In embodiments, the drive gear **471** can be driven to rotate on its axis by the motor **60**A directly or indirectly.

[0089] Optionally, the transmission assembly **462** further can include an intermediate gear **464** coupled to the first end plate **451**A by a bearing **473** (FIGS. **17** and **18**) received in a transmission guide **475** mounted on the first end plate **451**A. For example, the intermediate gear **464** can rotate about its axis on the bearing **473** and the transmission guide **475** can define a path **477** (e.g., a curved path as shown in FIGS. **10** and **14-18** or any suitable path) between the driven rollers **56**/**58**. Accordingly, in operation, the intermediate gear **464** can move/translate on its bearing **473** along the path **477** from a first end **477**A to a second end **477**B. In exemplary embodiments, the transmission guide **475** can be made of a self-lubricating plastic or any suitable material. The drive gear **471** can engage the intermediate gear **464**, wherein a plurality of teeth **471**A arranged along the circumference of the drive gear are configured to enmeshed with or otherwise engage a plurality of teeth **464**A arranged along the circumference of the intermediate gear **464**. It is contemplated that the curve of the path **477** in the transmission guide **475** can help retain the intermediate gear **464** in engagement with the drive gear **471** as the intermediate gear **464** translates along the path **477**.

[0090] As exemplarily illustrated in FIGS. **10** and **14-18**, the transmission assembly **462** further can include a first roller gear **467** mounted to a distal end of the first driven roller **56** and a second roller gear **469** mounted to a distal end of the second drive roller **58**. In embodiments, the roller gears **467**/**469** can be coupled to the respective driven rollers **56**/**58** (e.g., by respective axles extending through respective openings in the first end plate **451**A) so that rotation of a selected one of the roller gears **467**/**469** causes the respective driven roller **56**/**58** to rotate on its axis. As exemplarily shown, the first roller gear **467** can be mounted proximate the first end **477**A of the transmission guide **475** so that the first roller gear **467** engages the intermediate gear **464** (e.g., teeth **467**A arranged along the circumference of the first roller gear **467** engage the teeth **464**A of the intermediate gear **464**) when the intermediate gear **464** is at the first end **477**A of the transmission guide **475**. Similarly, the second roller gear **469** can be mounted proximate the second end **477**B of the transmission guide **475** so that the second roller gear **469** engages the intermediate gear **464** (e.g., teeth **469**A arranged along the circumference of the second roller gear **469** engage the teeth **464**A of the intermediate gear **464**) when the intermediate gear **464** is at the second end **477**B of the transmission guide **475**.

[0091] In operation, according to exemplary embodiments, the motor **60**A can rotate in the direction D**1** shown in FIG. **16**, rotating the drive gear **471** in the direction D**1**. The engagement between the intermediate gear **464** and the drive gear **471** can urge the intermediate gear **464** toward the first end **477** of the transmission guide **475** and in engagement with the first roller gear **467**. Accordingly, the rotation of the drive gear **471** is transmitted to the first roller gear **467** via the intermediate gear **464** so that the first driven roller **56** is rotated in the direction D**1** to draw the sheet material **11** between the first driven roller **56** and the guide rollers **442**A/**442**B along the discharge path toward and out from the discharge **15** of the dispenser housing **12**.

[0092] In embodiments, it is contemplated that the transmission assembly **462** can be operated to disengage the first driven roller **56** and engage the second driven roller **58**. For example, the controller **100** can reverse the direction of the motor **60**A (e.g., due to a signal from a supply sensor S**1** that the first supply of sheet material is depleted) so that the motor turns the drive gear **471** in the direction D**2** shown in FIG. **17**. In this aspect, the intermediate gear **464** is moved along the path **477** in the transmission guide **475** in the direction of the arrow A, away from the first roller gear **467** and the first end **477**A of the path **477** by the drive gear **471** until the intermediate gear **464** reaches the second end **477**B of the path **477** and engages the second roller gear **469**. As shown in FIG. **18**, continued rotation of the drive gear **471** causes the intermediate gear **464** to rotate on its axis at the second end **477**B, causing the second roller gear **469** to rotate. Accordingly, the rotation of the drive gear **471** is transmitted to the second roller gear **469** via the intermediate gear **464** so that the second driven roller **58** is rotated in the direction D**2** to draw the sheet material **11** between the second driven roller **58** and the guide rollers **444**A/**444**B along the discharge path toward and out from the discharge **15** of the dispenser housing **12**. The transmission assembly **462** can be operated to disengage the second driven roller **58** and engage the first driven roller **56** by reversing the above process (e.g., due to a signal produced by a supply sensor S**1** that indicates that the first supply of sheet material is depleted).

[0093] Additional views of the features of FIGS. **6-18** are shown in FIGS. **19**A-**19**E according to embodiments of the disclosure.

[0094] In embodiments, it is contemplated that the gear driven transmission assembly **462** generally can have a quieter operation than belt driven transmissions. Further, in various aspects, the gear driven transmission assembly **462** can be more reliable, such as by reducing or eliminating slipping between components and/or using more durable components. Further, the gear driven transmission assembly **462** can reduce or eliminate the need for separate biasing apparatus since rotation of the drive gear **471** urges the intermediate gear **464** toward the first roller gear **467** or the second roller gear **469** in addition to rotating the intermediate gear **464**.

[0095] FIGS. **20**A-**27**B show a dispensing system or mechanism **650** for a dispenser according to additional embodiments of the disclosure, which embodiments are generally similar to the prior embodiments except for variations noted and variations that will be apparent to one of ordinary skill in the art. Accordingly, similar or identical features of the embodiments have been given like or similar reference numbers. In the embodiments of FIGS. **20**A-**27**B, the dispensing system **450** includes a belt driven transmission assembly **662** with two belts **664**A, **664**B selectively driven by a centrally located gear clutch assembly **671** mounted on a drive mechanism **660**.

[0096] As exemplarily shown in FIGS. **21**A and **21**B, the belt driven transmission assembly **662** can be mounted at an end wall or end plate **651**A of the dispensing system **650** and can include the two drive belts **664**A, **664**B operatively connecting or engaging the drive mechanism **660** and the driven rollers **56**, **58** to transfer power therebetween for selectively driving rotation of the first driven roller **56** and/or the second driven roller **58**, which driven rollers **56**, **58** can be supported between the end plate **651**A and an opposing end wall or end plate **651**B. In embodiments, the gear clutch assembly **671** can be connected to the driven rollers **56**, **58** by the respective drive belts **664**A, **664**B, which can engage corresponding belt pulleys or belt gears **667**A, **667**B connected to the respective driven rollers **56**, **58**.
[0097] In additional optional aspects and as shown in FIGS. **23** and **25**A-**25**D, the drive mechanism **660** can include a motor **660**A (for example and without limitation, a brushless servo or stepper motor, or other, similar type of variable speed, reversible electric motor, or other suitable drive system) and a drive shaft **660**B. As shown in FIGS. **25**C and **25**D, the drive shaft **660**B can include a proximal section **681**A spaced from a distal section **681**B by a recess **681**C or other suitable dividing feature.
[0098] As described in more detail below, each of the respective proximal section **681**A and the distal section **681**B can be configured to operatively engage respective portions of the gear clutch assembly **671**. In some exemplary embodiments, a C-clip **684** can engage the recess **681**C and can extend between the portions of the gear clutch assembly **671** (FIGS. **22**, **23**, **25**A, and **25**B) to help keep the portions of the gear clutch assembly **671** separate from one another. The drive shaft **660**B could be otherwise configured without departing from the disclosure. For example, and without limitation, the recess **681**C and the C-clip **684** could be omitted and/or the drive shaft **660**B can be stepped so that the distal section **681**B has a different diameter than the proximal section **681**A (e.g., the diameter of the distal section **681**B could be stepped up from the diameter of the proximal section **681**A or the diameter of the distal section **681**B could be stepped down from the diameter of the proximal section **681**A) to help keep the portions of the gear clutch assembly **671** separate. In the illustrated exemplary embodiments, the drive motor **660**A can be mounted in a defined interior chamber of the dispenser housing and on an interior side of the end plate **651**A, and the drive shaft **660**B can extend through the end plate **651**A (e.g., via a through-bore, a clearance opening, a bushing, a bearing, and/or other suitable features) to engage the gear clutch assembly **671** on the outer side of the end plate **651**A. In other embodiments, the drive motor **660**A could be mounted on the exterior side of the end plate **651**A.
[0099] As shown in FIGS. **21**A-**23**, **25**A, and **25**B, the gear clutch assembly **671** can include a first or proximal drive pulley or drive gear **671**A engaging or mounted on the proximal section **681**A of the drive shaft **660**B and a second or distal drive pulley or drive gear **671**B engaging or mounted on the distal section **681**B of the drive shaft **660**B. As shown in FIGS. **21**A, **21**B, **27**A, and **27**B, the first drive belt **664**A engages the proximal drive gear **671**A and the first belt gear **667**A so that they are rotatably coupled together by the first drive belt, and the second drive belt **664**B engages the distal drive gear **671**B and the second belt gear **667**B so that they are rotatably coupled together by the second drive belt. In embodiments, the C-clip **684** can be positioned between the drive gears **671**A, **671**B in the recess **681**C of the drive shaft **660**B so that the drive gears **671**A, **671**B are spaced from one another by at least the C-clip **684**.
[0100] In exemplary embodiments, the gear clutch assembly **671** can include or incorporate one or more clutch assemblies or mechanisms **670** (FIG. **25**B), such as hybrid or one-way clutch mechanisms, that allow for selective transfer of power between the drive mechanism **660** and the driven rollers **56**, **58**. For example and as shown in FIGS. **22**, **23**, **25**A, and **25**B, the clutch mechanisms **670** can be incorporated or integrated with the respective drive gears **671**A, **671**B. Accordingly, in operation, when the drive mechanism **660** is driven in a first direction D**1** (FIGS. **21**B and **25**D), the clutch assembly **670** of the proximal drive gear **671**A will lock/engage for transfer of power/torque to the first driven roller **56** (e.g., via the first drive belt **664**A and the first

belt gear **667**A) so that the first driven roller **56** is driven by the drive mechanism **660** and rotated to dispense its corresponding supply of sheet material (while the clutch assembly **670** of the distal drive gear **671**B remains generally disengaged). In addition, when the drive mechanism **660** is driven in the opposite direction D**2** (FIGS. **21**B and **25**D), the clutch assembly **670** of the proximal drive gear **671**A will unlock or disengage such that there is no transfer of power/torque between the drive mechanism **660** and the first driven roller **56** (while the clutch assembly **670** for the distal drive gear **671**B engages or locks for transfer of power/torque to the second driven roller **58**, e.g., via the second drive belt **664**B and the second belt gear **667**B, so that the second driven roller **58** is rotated to dispense its corresponding supply of sheet material).

[0101] In one example construction, as generally illustrated in FIGS. **23**, **25**A, and **25**B, each clutch assembly **670** can include one or more tracks/races, such as an inner race **672** and an outer race **674**, that rotate together (when engaged) or independently of one another (when disengaged). In exemplary embodiments, the inner and outer races **672**, **674** of the clutch assemblies **670** can be configured in a similar or identical manner as the inner and outer races **72**, **74** of FIG. **2**B. Alternatively, one or both of the clutch assemblies **670** could be otherwise configured without departing from the scope of the application. In an exemplary embodiment, both of the inner races **672** of the drive gears **671**A, **671**B are turned by the drive shaft **660**B when the motor **660**A turns the drive shaft in either direction D**1**, D**2**. However, in operation, the clutch assemblies **670** are configured so that, when the drive shaft **660**B is turned in the direction D**1**, only the clutch assembly **670** of the proximal drive gear **671**A engages to cause the outer race **674** to turn with the inner race **672**, thereby moving the first drive belt **664**A, which turns the first belt gear **667**A and the first driven roller **56**. The clutch assembly **670** of the distal drive gear **671**B is disengaged so that the inner race **672** and the outer race **674** are able to rotate independently when the drive shaft **660**B is rotated in the direction D**1** and the outer race **674** does not rotate with the inner race **672** so that the second drive belt **664**B, the second belt gear **667**B, and the second driven roller **58** are not turned. Similarly, when the drive shaft **660**B is operationally turned in the direction D**2**, only the clutch assembly **670** of the distal drive gear **671**B engages to cause the outer race **674** to turn with the inner race **672**, thereby moving the second drive belt **664**B, which turns the second belt gear **667**B and the second driven roller **58**. In this aspect, the clutch assembly **670** of the proximal drive gear **671**A is disengaged so that the inner race **672** and the outer race **674** are able to rotate independently when the drive shaft **660**B is rotated in the direction D**2** and the outer race **674** does not rotate with the inner race **672** so that the first drive belt **664**A, the first belt gear **667**A, and the first driven roller **56** are not turned.

[0102] In the illustrated embodiments, the proximal and distal drive gears **671**A, **671**B can be similar or identical to one another and can be mounted on the drive shaft **660**B in opposite orientations (FIGS. **22**, **23**, **25**A, and **25**B). Alternatively, in other exemplary embodiments, the drive gears **671**A, **671**B could be configured differently.

[0103] As shown in FIGS. **21**A-**22**, **24**, **26**A, and **26**B, the belt gears **667**A, **667**B are mounted to respective drive shafts or roller extensions **668** of the respective driven rollers **56**, **58**. As shown in FIG. **22**, each of the roller extensions **668** includes a cylindrical portion **668**A that is supported in a respective opening **686** of the end plate **651**A by a bearing **686**A and/or a bushing **686**B (e.g., the cylindrical portion **668**A engages an inner race of the bearing **686**A or an inner surface of the bushing **686**B). In optional embodiments, a distal section **668**B of the roller extension **668** can be configured to mate with or otherwise engage a central bore **687** of the respective belt gear **667**A, **667**B. For example, in an optional aspect, the distal section **668**B can have a flat surface **668**C on either side of the roller extension **668**, and the flat surfaces **668**C can engage respective flat surfaces on the interior surface of the belt gear extending along the central bore **687**. Accordingly, operable rotation of the belt gears **667**A, **667**B can cause the respective roller extension **668** to turn, thereby turning the respective driven roller **56**, **58**.

[0104] As shown in FIGS. **27**A and **27**B, the belt gears **667**A, **667**B can be similar or identical to one another and can be mounted on the respective roller extensions **668** in opposite orientations in order to align the belt engaging portions of the belt gears **667**A, **667**B with the respective drive gears **671**A, **671**B. For example and without limitation, each of the belt gears **667**A, **667**B can include a spacer portion **688**. In this aspect, it is contemplated that the second belt gear **667**B can be oriented so that the spacer portion **688** is positioned between the belt engaging portion of the belt gear and the end plate **651**A, and the spacer portion **688** of the first belt gear **667**A is spaced from the end plate **651**A by the belt engaging portion of the belt gear. The belt gears **667**A, **667**B could be otherwise configured and/or could be optionally configured different from one another (e.g., the spacer portion **688** could be omitted on the first belt gear **667**A).

[0105] In embodiments, the transmission assembly **662** of FIGS. **20**A-**27**B can be easier to assemble due to the central location of the gear clutch assembly **671**. In addition, the belt gears **667**A, **667**B, located at the outer portions of the transmission assembly, can be desirably made of lighter, cheaper materials (e.g., plastic instead of metal). In embodiments, the configuration of the transmission assembly **662** can be quieter and can increase the efficiency of the dispensing system **650**.

[0106] FIGS. **28**A and **28**B show views of other optional embodiments of a dispensing system **650′**, and FIGS. **28**C and **28**D show views of other optional embodiments of a drive mechanism **660′**.

[0107] In one exemplary embodiment, it is contemplated that the sheet material dispenser can include a dispenser housing with a first and a second supply of sheet material supported therein and including a first driven roller rotatably mounted within the dispenser housing that is configured to drive sheet material from the first supply of sheet material and a second driven roller rotatably mounted within the dispenser housing that is configured to drive sheet material from the second supply of sheet material. In this aspect, the sheet material dispenser further can include a drive mechanism operably coupled to a select one of the first or second driven roller to selectively drive rotation of the select one of the first or the second driven roller. In this exemplary aspect, it is contemplated that the drive mechanism comprises a transmission assembly configured to selectively drive rotation of the select one of the first or the second driven roller; and a motor configured to drive in a first rotation direction to cause the transmission assembly to disengage with the second driven roller and engage and rotate the first driven roller to affect the movement of the sheet material from the first supply of sheet material along a first discharge path toward and out from a discharge defined in the dispenser housing. In this aspect, the motor is further configured to drive in a second rotation direction, opposite to the first rotation direction, to cause the transmission assembly to disengage with the first driven roller and to engage and rotate the second driven roller to affect the movement of sheet material from the second supply of sheet material along a second discharge path toward and out from the discharge of the dispenser housing. The sheet material dispenser can further include a control circuitry in communication with the drive mechanism that is programmed to selectively actuate the motor in the first or second rotative direction to feed the sheet material from the selected supply of sheet material through the discharge defined in the dispenser housing.

[0108] In a further exemplary aspect, the transmission assembly of the sheet material dispenser can include a plurality of gears for transferring power from the drive mechanism to a select one of the first or the second driven roller. For example, and without limitation, the plurality of gears can comprise a drive gear, an intermediate gear, a first roller gear, and a second roller gear. In this exemplary aspect, the drive gear is mounted to the motor and the intermediate gear is configured to be slidably received within a transmission guide defined in the first end plate of the housing that extends along a path between the respective first and second driven rollers. As shown, the intermediate gear is positioned in engagement with the drive gear and rotates in a direction that is in opposition to the select first or second rotation direction of the motor.

[0109] Further, the first roller gear is mounted to a distal end of the first driven roller and the second roller gear is mounted to a distal end of the second drive roller. As shown, the selective rotation of a respective first or second roller gear affects a complementary rotation of the coupled first or second driven roller. More particularly, in this exemplary aspect, the first roller gear is mounted proximate a first end of the transmission guide so that the first roller gear is configured to engage the intermediate gear when the intermediate gear is at the first end of the transmission guide. Similarly, the second roller gear is mounted proximate a second end of the transmission guide so that the second roller gear is configured to engage the intermediate gear when the intermediate gear is at the second end of the transmission guide.

[0110] In another aspect, the sheet material dispenser can further include a first guide roller and a second guide roller. In this aspect, the dispenser housing has a first end plate and a second, spaced and opposed, end plate and, as shown, each respective end plate can define a slot. It is contemplated that the first and second guide rollers can be configured for mounting therein the respective slots of the first and second end plates and can be further configured for biased movement along the respective slots of the first and second end plates such that the respective first and second guide rollers are urged toward the respective first and second driven rollers. In this exemplary aspect, each of the first and second guide rollers can include a pair of bearing assemblies positioned so that one bearing assembly of the pair of bearing assemblies can be positioned at a proximal end of the respective guide roller and another bearing assembly of the pair of bearing assemblies can be positioned at a distal end of the respective guide roller.

[0111] Still further, it is contemplated in this aspect that the sheet material dispenser can include a first biasing member mounted within the dispenser housing that is configured to urge the first guide roller along the slots and toward the first driven roller and a second biasing member mounted within the dispenser housing configured to urge the second guide roller along the slots and toward the second driven roller.

[0112] In another exemplary embodiment, it is contemplated that the sheet material dispenser can include a dispenser housing, a first driven roller, a second driven roller, a drive mechanism, and a control circuity. In this aspect, the dispenser housing has a first and a second supply of sheet material supported therein the housing. Further, the first driven roller is configured to be rotatably mounted within the dispenser housing to drive sheet material from the first supply of sheet material and, similarly, the second driven roller is configured to be rotatably mounted within the dispenser housing to drive sheet material from the second supply of sheet material.

[0113] As exemplarily illustrated, the drive mechanism is operably coupled to a select one of the first or second driven roller to selectively drive rotation of the select one of the first or the second driven roller. In this exemplary aspect, the drive mechanism comprises a motor and a plurality of gears. The motor configured to drive in a first rotation direction and further configured to drive in a second rotation direction, opposite to the first rotation direction. For example, and without limitation, the plurality of gears for transferring power from the drive mechanism to the select one of the first or the second driven roller includes a drive gear, an intermediate gear, a first roller gear and a second roller gear. In this exemplary aspect, the drive gear is mounted to the motor and the intermediate gear is configured to be slidably received within a transmission guide defined in the first end plate of the housing that extends along a path between the respective first and second driven rollers. As shown, the intermediate gear is positioned in engagement with the drive gear and rotates in a direction that is in opposition to the select first or second rotation direction of the motor.

[0114] The exemplary control circuitry is in communication with the drive mechanism and is programmed to selectively actuate the motor in the first or second rotative direction to feed the sheet material from the selected supply of sheet material through the discharge defined in the dispenser housing.

[0115] In this exemplary aspect, when the motor is selectively driven in the first rotation direction, the intermediate gear engages with the first roller gear and disengages from the second roller gear to affect the rotation of the first driven roller and to affect movement of sheet material from the first supply of sheet material along a first discharge path toward and out from the discharge. Similarly, when the motor is selectively driven in the second rotation direction, the intermediate gear disengages with the first roller gear and engages with the second roller gear to affect the rotation of the second driven roller and to affect movement of sheet material from the second supply of sheet material along a second discharge path toward and out from the discharge of the dispenser housing. More particularly, in this exemplary aspect, it is contemplated that the first roller gear is mounted proximate a first end of the transmission guide so that the first roller gear is configured to engage the intermediate gear when the intermediate gear is at the first end of the transmission guide, Similarly, it is contemplated that the second roller gear is mounted proximate a second end of the transmission guide so that the second roller gear is configured to engage the intermediate gear when the intermediate gear is at the second end of the transmission guide.

[0116] In a further exemplary embodiment, it is contemplated that the sheet material dispenser can include a dispenser housing, a first driven roller, a second driven roller, and a drive mechanism operably coupled the first and second driven rollers. In this aspect, the dispenser housing has a first and a second supply of sheet material supported therein. Further, the first driven roller is configured to be rotatably mounted within the dispenser housing to drive sheet material from the first supply of sheet material and the second driven roller is configured to be rotatably mounted within the dispenser housing to drive sheet material from the second supply of sheet material. In this exemplary aspect, the sheet material dispenser can further include a control circuitry in communication with the drive mechanism that is programmed to selectively actuate the motor in the first or second rotative direction to feed the sheet material from the selected supply of sheet material through the discharge defined in the dispenser housing.

[0117] In this exemplary aspect and as illustrated, the drive mechanism can include a motor, a first and second belt gear, a first and second drive gear, and respective first and second drive belts, and a plurality of clutch assemblies. In this aspect, the motor is configured to drive in a first rotation direction and in a second, opposite, rotation direction and further has a drive shaft that has a proximal section and a spaced distal section.

[0118] In further exemplary aspects, the first belt gear is coupled to the first driven roller and the second belt gear is coupled to the second driven roller. Further, as shown, the first drive gear is mounted on the proximal section of the drive shaft and the second drive gear is mounted on the distal section of the drive shaft. As further shown, the first drive belt is configured to engage the first belt gear and the first drive gear and similarly, the second drive belt is configured to engage the second belt gear and the second drive gear.

[0119] At least one clutch assembly of the plurality of clutch assemblies is configured to be operably integrated with the first drive gear and at least one clutch assembly configured to be operably integrated with the second drive gear. In this aspect, upon selective rotation of the motor in the first rotation direction, the at least one clutch assembly that is configured to be operably integrated with the first drive gear disengages from the second drive gear and operatively engages the first drive gear to affect transfer of torque to the first drive gear so that the first driven roller is driven by the first drive belt and first drive gear to rotate and dispense its supply of sheet material. Similarly, upon selective rotation of the motor in the second rotation direction, the at least one clutch assembly that is configured to be operably integrated with the second drive gear disengages from the first drive gear and operatively engages the second drive gear to affect transfer of torque to the second drive gear so that the first driven roller is driven by the second drive belt and second drive gear to rotate and dispense its supply of sheet material.

[0120] Exemplarily, each clutch assembly can include an inner race and an outer race that are configured to rotate together when the clutch assembly is engaged and are configured to act independently of one another when the clutch assembly is disengaged. In operation, when the drive shaft is turned in the first rotative direction, each clutch assembly is configured so that the at least one clutch assembly integrated with the first drive gear engages to cause the respective clutch assembly outer race to turn with the inner race, affecting movement of the first drive belt and rotation of the first belt gear and the first driven roller. Additionally, when the drive shaft is turned in the first rotative direction, the at least one clutch assembly integrated with the second drive gear is disengaged so that the respective clutch assembly inner race and the outer race rotate independently of each other such that the outer race does not rotate with the inner race and no movement is affected on the second drive belt.

[0121] In a similar embodiment in operation, when the drive shaft is turned in the second rotative direction, each clutch assembly is configured so that only the at least one clutch assembly integrated with the second drive gear engages to cause the respective clutch assembly outer race to turn with the inner race, affecting movement of the second drive belt and rotation of the second belt gear and the second driven roller. Thus, when the drive shaft is turned in the second rotative direction, the at least one clutch assembly integrated with the first drive gear is disengaged so that the respective clutch assembly inner race and the outer race rotate independently of each other such that the outer race does not rotate with the inner race and no movement is affected on the first drive belt.

[0122] Any of the features of the various embodiments of the disclosure can be combined with, replaced by, or otherwise configured with other features of other embodiments of the disclosure without departing from the scope of this disclosure.

[0123] The foregoing description generally illustrates and describes various embodiments of the present invention. It will, however, be understood by those skilled in the art that various changes and modifications can be made to the above-discussed construction of the present invention without departing from the spirit and scope of the invention as disclosed herein, and that it is intended that all matter contained in the above description or shown in the accompanying drawings shall be interpreted as being illustrative, and not to be taken in a limiting sense. Furthermore, the scope of the present disclosure shall be construed to cover various modifications, combinations, additions, alterations, etc., above and to the above-described embodiments, which shall be considered to be within the scope of the present invention. Accordingly, various features and characteristics of the present invention as discussed herein may be selectively interchanged and applied to other illustrated and non-illustrated embodiments of the invention, and numerous variations, modifications, and additions further can be made thereto without departing from the spirit and scope of the present invention as set forth in the appended claims.

Claims

1. A sheet material dispenser comprising: at least one driven roller configured to feed sheet material from one or more supplies of sheet material; a transmission assembly; and a motor coupled to the transmission assembly and configured to operate in a first rotation direction to cause the transmission assembly to engage the at least one driven roller to feed the sheet material from a first supply of sheet material along a first discharge path toward a discharge, and operate in a second rotation direction, opposite to the first rotation direction, to cause the transmission assembly to engage the at least one driven roller to feed the sheet material from a second supply of sheet material along a second discharge path toward the discharge; and a control system in communication with the motor, the control system including programming configured to selectively actuate the motor in the first rotation direction or the second rotation direction for selectively feeding the sheet material from a selected one of the supplies of the sheet material toward the discharge.

2. The sheet material dispenser of claim 1, wherein the at least one driven roller includes a first driven roller configured to feed sheet material from the first supply of sheet material and a second driven roller configured to feed sheet material from the second supply of sheet material.

3. The sheet material dispenser of claim 2, wherein the transmission assembly comprises a plurality of gears configured to transfer power or torque from the motor to one of the first driven roller or the second driven roller.

4. The sheet material dispenser of claim 2, wherein the transmission assembly comprises: a drive gear coupled to the motor; a first roller gear coupled to the first driven roller; a second roller gear coupled to the second drive roller; and an intermediate gear configured to be driven by the drive gear and to selectively engage the first roller gear or the second roller gear; and wherein the driving of the intermediate gear by the drive gear causes a complementary rotation of the first driven roller when the intermediate gear is engaged with the first roller gear, and a complementary rotation of the second driven roller when the intermediate gear is engaged with the second roller gear.

5. The sheet material dispenser of claim 4, wherein the transmission assembly further comprises a transmission guide; and wherein the intermediate gear is movable along the transmission guide between a position in engagement with the first roller gear for driving rotation of the first driven roller, and a position in engagement with the second roller gear for driving rotation of the second driven roller.

6. The sheet material dispenser of claim 2, further comprising a first guide roller positioned adjacent the first driven roller and a second guide roller positioned adjacent the second driven roller; and wherein the first and second guide rollers are urged toward the first and second driven rollers so as to engage the sheet material therebetween.

7. The sheet material dispenser of claim 1, wherein the at least one driven roller comprises a first driven roller and a second driven roller; and wherein each of the first driven roller and the second driven roller includes an integrated clutch assembly configured to enable a selective transfer of power or torque between the motor and the first driven roller and the second driven roller.

8. A dispenser, comprising: a first driven roller configured to feed sheet material from a first supply of sheet material; a second driven roller configured to feed sheet material from a second supply of sheet material; a motor configured to operate in a first rotation direction and further configured to operate in a second rotation direction, opposite to the first rotation direction; a transmission assembly configured to selectively transfer power or torque from the motor to the first driven roller or the second driven roller; and a control system in communication with the motor and configured to selectively cause the motor to operate in the first rotation direction so as to drive rotation of the first driven roller for feeding the sheet material from the first supply of sheet material, and cause the motor to operate in the second rotation direction so as to drive rotation of the second driven roller for feeding the sheet material from the second supply of sheet material.

9. The dispenser of claim 8, wherein the transmission assembly comprises: a plurality of gears configured for transferring power or torque from the motor to the first driven roller or the second driven roller, the plurality of gears comprising: a drive gear coupled to the motor; a first roller gear coupled to the first driven roller; a second roller gear coupled to the second driven roller; and an intermediate gear positioned in engagement with the drive gear so as to be rotated by the drive gear as the motor is operated in the first rotation direction and the second rotation direction; and wherein the intermediate gear is configured to move between engagement with the first roller gear and the second roller gear for selectively rotating the first roller gear and the second roller gear.

10. The dispenser of claim 9, wherein when the motor is operated in the first rotation direction the intermediate gear is caused to engage with the first roller gear and disengage from the second roller gear to drive rotation of the first driven roller for feeding the sheet material from the first supply of sheet material along a first discharge path; and wherein when the motor is operated in the second rotation direction the intermediate gear is caused to disengage from the first roller gear and engage with the second roller gear to drive rotation of the second driven roller for feeding the sheet material from the second supply of sheet material along a second discharge path.

11. The dispenser of claim 8, wherein each of the first driven roller and the second driven roller includes an integrated clutch assembly configured to enable a selective transfer of power or torque between the motor and the first driven roller and the second driven roller.

12. The dispenser of claim 8, further comprising a sheet material detection sensor in communication with the control system, the sheet material detection sensor configured to provide one or more signals to the control system, including an indication that the sheet material dispensed has been removed from the dispenser, an indication that the sheet material cannot be dispensed from the first or second supplies of sheet material, an indication of a supply level of the sheet material of at least one of the first and second supplies of sheet material, or combinations thereof.

13. The dispenser of claim 8, further comprising a plurality of clutch assemblies, at least one clutch assembly configured to facilitate rotation of the first driven roller and at least one clutch assembly configured to facilitate rotation of the second driven roller; wherein, upon operation of the motor in the first rotation direction, the at least one clutch assembly configured to facilitate rotation of the first driven roller is engaged to transfer power or torque to the first driven roller to dispense the sheet material from the first supply of sheet material; and wherein, upon operation of the motor in the second rotation direction, the at least one clutch assembly configured to facilitate rotation of the second driven roller is engaged to transfer power or torque to the second driven roller to dispense the sheet material from the second supply of sheet material.

14. The dispenser of claim 8, further comprising a monitoring system in communication with the control system; wherein the monitoring system includes a series of sensors located adjacent the first and second driven rollers and configured to monitor rotation thereof; and wherein the control system is further configured to determine a remaining amount of sheet material of at least one of the first and second supplies of sheet material based on a monitored number of rotations of the first or second driven rollers and a length of sheet material dispensed during a dispensing operation.

15. A dispenser for dispensing sheet material, comprising: a first supply of sheet material and a second supply of sheet material; a first driven roller configured to feed the sheet material from the first supply of sheet material; a second driven roller configured to feed the sheet material from the second supply of sheet material; a drive mechanism operably coupled to the first and second driven rollers and configured to selectively drive rotation of the first driven roller and the second driven roller; wherein the drive mechanism comprises: a motor configured to operate in a first rotation direction and in a second rotation direction; and a plurality of clutch assemblies, at least one clutch assembly configured to be selectively engaged for driving rotation of the first driven roller, and at least one clutch assembly configured to be selectively engaged for driving rotation of the second driven roller; wherein, when the motor is driven in the first rotation direction, the at least one clutch assembly configured to be selectively engaged for driving rotation of the first driven roller in rotation direction is caused to engage and drive rotation of the first driven roller to dispense the sheet material from the first supply of sheet material while the at least one clutch assembly configured to be selectively engaged for driving rotation of the second driven roller is disengaged; and wherein, when rotation of the motor is driven in the second rotation direction, the at least one clutch assembly configured to be selectively engaged for driving rotation of the second driven roller is caused to engage and drive rotation of the second driven roller to dispense the sheet material from the second supply of sheet material while the at least one clutch assembly configured to be selectively engaged for driving rotation of the first driven roller is disengaged.

16. The dispenser of claim 15, further comprising a control system in communication with the drive mechanism, the control system including a control circuitry configured to selectively actuate the motor to operate in the first rotation direction or the second rotation direction to feed the sheet material from a selected one of the first and second supplies of sheet material to a discharge of the dispenser.

17. The dispenser of claim 16, further comprising a monitoring system in communication with the control system; wherein the monitoring system includes a series of sensors located adjacent the first and second driven rollers and configured to monitor rotation thereof; and wherein the control system is further configured to determine a remaining amount of sheet material of at least one of the first and second supplies of sheet material based on a monitored number of rotations of the first or second driven rollers and a length of sheet material dispensed during a dispensing operation.

18. The dispenser of claim 16, further comprising a sheet material detection sensor in communication with the control system and configured to provide one or more signals to the control system; wherein the one or more signals include an indication that the sheet material dispensed has been removed from the dispenser, an indication that the sheet material cannot be dispensed from the first or second supplies of sheet material, an indication of a supply level of the sheet material of at least one of the first and second supplies of sheet material, or combinations thereof.

19. The dispenser of claim 15, wherein the drive mechanism further comprises: a first belt gear coupled to the first driven roller; a second belt gear coupled to the second driven roller; a first drive gear positioned along a drive shaft of the motor and a second drive gear along the drive shaft of the motor; a first drive belt configured to engage the first belt gear and the first drive gear for driving the rotation of the first driven roller; and a second drive belt configured to engage the second belt gear and the second drive gear for driving the rotation of the second driven roller.

20. The dispenser of claim 19, wherein, upon operation of the motor in the first rotation direction, the at least one clutch assembly configured to be selectively engaged for driving rotation of the first driven roller is engaged with the first drive gear to dispense the sheet material from the first supply of sheet material; and wherein, upon operation of the motor in the second rotation direction, the at least one clutch assembly configured to be selectively engaged for driving rotation of the second driven roller is engaged with the second drive gear to drive rotation of the second driven roller to dispense the sheet material from the second supply of sheet material.